

For your final project, you will demonstrate your mastery of the course material by applying it to a dataset of your choice. You will have the opportunity to explore and analyze the data, design and implement a database schema, write complex queries, and present your findings in a compelling way. The project is designed to be open-ended to allow you to pursue your interests and creativity, while also ensuring that you apply the techniques and concepts you have learned throughout the course.

This is also an opportunity to produce something you can point to in your portfolio when applying to jobs. The strongest projects come from consistent, regular work over several weeks – not from a last-minute push. Start brainstorming early.

See Canvas for deadlines. Lightning talks will be held the last week of class.

1 Assignments

The final project is organized around three **components** – a dataset (Component 1), an application built on top of it (Component 2), and a set of concrete outputs you produce (Component 3) – each detailed in the **Project Components** section below. The components describe *what you build*; the table below describes *what you turn in*.

There are three graded Canvas assignments associated with this project. Each is graded ESNU against a rubric defined later in this document.

Assignment	What you submit	How it is graded
Project Proposal	PDF proposal (group members, dataset, application, anticipated outputs).	Instructor grades against the proposal rubric (§3.1).
Artifact Self-Assessment	Filled-in ESNU template for Components 1, 2, and chosen Component 3 outputs, with supporting artifacts.	Instructor regrades against the same rubrics; this is the grade on your project work itself.
Lightning Talk	≈5-minute group presentation, depending on the number of groups, with enough evidence for peers to score the same rubrics live.	Peer-graded against the same rubrics, combined with instructor review.

2 Groups

For this project, you must work in groups of 2 - 4. **No submissions will be accepted from individuals or from groups larger than 4.** As soon as possible, look through the various options for this project, decide on the type of project you are interested in, and start looking for potential group mates. Posting on Ed Discussion is a great way to find a like-minded group! **Once you have your group membership figured out, assign yourself to an unused group in Canvas.**

While this is a group project, you will also be graded on your individual contributions to the project. Ensure that every group member has a voice in decision making and an opportunity to contribute in equal measure to the final product! If you experience any difficulty working on the project or with your group, *let your instructor know as soon as possible*. You are responsible for your individual contributions to the project.

3 Project Proposal

Some weeks prior to the end of semester, your group must submit a final project proposal. You should read ahead in this document to understand the requirements of the final project before developing your proposal.

Your group proposal consists of a PDF document providing the following information:

1. A list of group members

2. A paragraph describing your proposed dataset. Briefly explain how you believe this dataset fulfills the requirements for Component 1 below.
3. A paragraph describing your proposed application (Component 2) and one or two outputs (Component 3) you anticipate including. Please be fairly specific here. For example, if you choose the “Data science and analysis” application, do not just state that you will “perform analysis and create visualizations” on the dataset; state what questions you hope to answer with your analysis and how your choice of visualization addresses these questions.

Note that this is just a **proposal**. It is perfectly fine for any of the above to change between the submission of your proposal and your actual final project (including your group membership). You should, however, demonstrate clearly that you have taken the time to understand your proposed dataset and how it can be used. If I feel that your proposal is insufficient, you will receive an N and be given the opportunity to revise and resubmit it.

3.1 Grading

The project proposal is graded ESNU.

Grade	Description
E	All S-level requirements met, with exceptional thoughtfulness: proposal clearly demonstrates deep engagement with the dataset (sample queries run, initial exploration done) and specifies concrete questions or user stories that go beyond generic goals.
S	Contains all required elements: group members listed, dataset described with explanation of how it meets Component 1, and a specific description of the proposed application (Component 2) and anticipated outputs (Component 3). Dataset and plan are plausible and show evidence of preliminary investigation.
N	Required elements present but descriptions are vague, generic (“we will perform analysis and create visualizations”), or do not demonstrate that the group has examined the dataset.
U	Missing one or more required elements (group members, dataset description, application, or outputs).

3.2 Token Usage for the Project Proposal

Because of the time-sensitive nature of this proposal, you may use a token (one for each group member!) to submit the proposal up to one week late. No proposals will be accepted more than one week late.

4 Project Components

Each project component has options to provide you with some flexibility in designing a project that matches your interests and that makes sense for the data you choose to work with. Read each description carefully to ensure you understand the requirements for each component. The ESNU rubrics defined in each component’s Grading subsection are what you will self-assess against in the Artifact Self-Assessment, and what the class will peer-grade against during your Lightning Talk.

4.1 Component 1: Dataset(s)

To fulfill the requirements for this component, choose a data set matching **one or more** of the descriptions below. In your Artifact Self-Assessment, state which option(s) you selected and how your data set meets the requirement. Provide links to your data sources and describe any limitations on use, i.e., if the data set is in the public domain or is restricted to non-commercial use, etc.

- **A large dataset.** “Large” is somewhat subjective, and may mean more than one thing. A data set with 100,000 records is definitely large. A data set with 10,000 rows may be “large” if it has many columns or is composed of multiple tables. A data set may be large if it presents some difficulty in loading, cleaning, or querying simply due to its size.

If you go this route, please keep it somewhat reasonable – our RDBMS is a shared resource! As a rule of thumb, keep it below 4 GB unless you want to work with it in a PostgreSQL instance on your own machines. You may work with a subset of a larger dataset if desired – just be sure to explain the process by which you select your subset in your Artifact Self-Assessment.

This option is likely to be a good fit with the **Data science and analysis** or **Advanced technology** options for Component 2, but of course may be used with other options.

- A data set that is composed of, or may be decomposed into, **3 or more interrelated tables**. A join of these tables must result in 2,000 or more rows. At least 2 tables must represent entities with significant structure – i.e., entities with multiple attributes of interest as opposed to “lookup tables”:
 - An example of an entity with significant structure might be historical meteorological statistics for agricultural districts in the state of Colorado recording rain accumulation, snow accumulation, average wind speeds, and so forth by year and month over multiple years.
 - An example of a lookup table that would not be considered an entity with significant structure is a table mapping state abbreviations (e.g., “CO”) to state names.

This option is likely to be a good fit with the **Software development** option for Component 2, but of course may be used with other options.

- **A “mash up” data set.** This is a dataset that is composed of 2 or more tables that are not obviously interrelated, but which can be combined by joining along some shared attribute either in the tables or that can be derived from attributes in the tables. The goal with this data set is to discover interesting analyses that offer some potential new insight into a topic. For example, you might combine a data set on alcohol sales by geographic location with a data set on climatic conditions in those locations to determine if factors such as temperature, inclement weather, etc. are driving factors for alcohol consumption.

This option is likely to be a good fit with the **Data science and analysis** option for Component 2, but of course may be used with other options.

4.1.1 Grading

Component 1 is graded on the following rubric:

Grade	Description
E	All S-level requirements met, and the dataset itself is unusually interesting: a unique or unusual source, a creative combination of multiple datasets, or something the instructor has not seen submitted before. The choice should feel like a deliberate find, not the first dataset that matched the requirement.
S	Provides a specific citation of the dataset (including access date and any usage restrictions), and a clear explanation of which option(s) were selected and how the dataset meets the stated requirement (record counts, table structure, joinability, etc. are concretely demonstrated).
N	Citation present but incomplete (missing access date, unclear licensing), or explanation of how the dataset meets the option is vague or not substantiated with specifics (row counts, column descriptions, etc.).
U	Missing citation, missing explanation, or dataset clearly does not meet the requirements of the selected option.

4.2 Component 2: Application

To fulfill the requirements for this component, apply concepts from this course to **one or more** of the following. In your Artifact Self-Assessment, state which option(s) you selected and a brief overview of your work and how it met the requirement for your option(s).

- **Data science and analysis.** Your project should dissect and analyze your data set in multiple ways (see **Outputs** section below). Your analysis must do more than scratch the surface with simple statistics or correlations – your goal is to extract insights from the data that are not trivially obvious

from an inspection of the data. While your work must include significant applications of SQL for data cleaning, aggregation, or other transformations, you may also utilize tools such as pandas, scikit-learn, TensorFlow, etc. It is acceptable for your work to be somewhat inconclusive if the methodology is **thorough** and correct.

- **Software development.** You will need to create application software (e.g., a web application, CLI application, or similar) that connects to the database and includes functionality to query, add, modify, and remove data. Your software must follow best practices with regards to database programming (e.g., prevent SQL injection attacks). Your application should connect to the course database unless you are combining this with the next option, but otherwise may use any language or frameworks you wish.
- **Advanced technology.** Use a new, unusual, or cutting-edge database system and provide a compelling demonstration of the advantages or trade-offs of using that database versus a classic RDBMS like PostgreSQL, MySQL, and so forth. Examples of an “advanced” database system include:
 - Graph databases such as Kuzu, Neo4j, Virtuoso, Blazegraph, or XTDB
 - Array databases such as SciDB
 - Distributed databases such as CockroachDB, FoundationDB, or TiDB
 - Vector databases such as Chroma, Pinecone, Qdrant, or pgvector
 - Streaming databases such as Apache Druid, Flink, and Materialize
 - Embedded analytical databases such as DuckDB

You are not limited to this list, but you should choose something significantly different from “out-of-the-box” PostgreSQL.

4.2.1 Grading

Component 2 is graded on the following rubric:

Grade	Description
E	All S-level requirements met, with an application that is unusually ambitious, polished, or original: the scope, design choices, or integration of course material goes clearly beyond what is needed to meet the requirement.
S	The Artifact Self-Assessment provides a clear and concise summary of the overall project, states which option(s) were selected, and explains specifically how the work meets the requirements of the chosen application (e.g., for Data science and analysis: the questions asked and methods used; for Software development: the functionality implemented and how it uses the DB; for Advanced technology: what the system is and what is being demonstrated).
N	Summary is present but vague or missing details; explanation of how the work meets the option is incomplete (e.g., claims “analysis” without describing what was analyzed, or claims “software” without describing its functionality).
U	Missing summary, missing explanation of how the work meets the chosen option, or the work does not actually meet the requirements of the selected option.

4.3 Component 3: Outputs

To fulfill the requirements for this component, include at least 4 of these deliverables. In your Artifact Self-Assessment, include a section listing which deliverables you have included (in addition to the supporting artifacts implementing each deliverable).

- **Data loading and cleaning.** Only include this deliverable if your data needed more than trivial cleaning. In your Artifact Self-Assessment, include a concise and complete description of the issues with your data set and the steps you took to correct them. Include a SQL script showing your data cleaning queries as a supporting artifact (properly formatted with a fixed-width font in your submission text, not a screenshot!).

Grade	Description
E	All S-level requirements met, and the cleaning goes clearly above and beyond: leverages domain knowledge to repair the data, performs non-trivial data engineering to get messy or semi-structured sources into a clean relational form, or builds a reproducible pipeline that handles cases a casual inspection would miss.
S	Performs at least 2 <i>non-trivial</i> cleaning operations. Provides clear before/after examples for each and justifies why the cleaning was required <i>for the specific uses of the dataset in this project</i> . Trivial operations (removing nulls, duplicate elimination, a plain cast between compatible types) do <i>not</i> count toward the 2. Examples of non-trivial cleaning that <i>do</i> count: parsing or extracting values from semi-structured fields (including type coercion that requires parsing), reconciling inconsistent categorical labels, unit normalization, outlier detection with justification, entity resolution across sources.
N	Fewer than 2 non-trivial cleaning operations, or the cleaning performed is trivial (duplicate/null/type), or justification is not tied to how the data is actually used in the project.
U	No meaningful cleaning identified, or cleaning does not affect data quality for the project's use case.

- **Database schema design.** Provide a clear and complete schema diagram of your database design, showing all tables, primary and foreign keys, and relationships. In your Artifact Self-Assessment, explain your design choices, such as normalization decisions and any trade-offs made between normalization and performance. Your documentation must include information about constraints implemented to maintain data integrity. If your database design evolved during the project, briefly explain the significant changes and why they were necessary. **Do not select this option if you only have a single table, or your tables have no relationships or constraints.**

Grade	Description
E	All S-level requirements met, with exceptional clarity or depth: e.g., quantitative evidence for normalization/performance tradeoffs, thoughtful constraint design beyond the minimum, or a schema evolution narrative that reflects real engineering judgment.
S	Includes a figure showing all tables with primary and foreign keys and relationships. States the normalization level reached (e.g., denormalized, 3NF, BCNF) and justifies why that level was chosen; the schema must actually satisfy the claimed NF (e.g., if BCNF is claimed, the relevant functional dependencies must be documented). Describes potential performance tradeoffs of the chosen normalization level. Documents any significant schema evolution during the project and why each change was made.
N	Diagram is present but messy, incomplete, or inaccurate; or normalization level is stated without justification / does not actually hold; or performance tradeoffs and schema evolution are not discussed.
U	Missing diagram, no discussion of normalization, or no relationships/constraints of consequence.

- **Interesting or complex queries.** Your query or set of queries should demonstrate multiple SQL techniques beyond simple SELECT...WHERE..., e.g., subqueries, CTEs, window functions, and grouping and aggregation. These queries must not be created simply to demonstrate techniques, but must have a recognizable function in your analysis, software, or evaluation. In your Artifact Self-Assessment, include an entry **for each query** providing the SQL (properly formatted with a fixed-width font in your submission text, not a screenshot!) and describing the design and purpose of the query.

Grade	Description
E	All S-level requirements met, and the queries stand out: they answer unusually interesting or non-obvious questions, demonstrate sophisticated use of advanced SQL features beyond what was covered in class, or are large/complex queries that accomplish something genuinely non-trivial (not just longer for its own sake).
S	At least 2 queries. Each query uses at least 2 instances of advanced SQL features (sub-queries, CTEs, window functions, grouping and aggregation). For each query, clearly describe what it computes and why it was written – i.e., its role in the project. Acceptable roles include: (1) a view or materialized view that encodes a reusable transformation the application or analysis depends on, with justification for why it is defined as a view; (2) a dataset-creation query that produces a non-trivial transformation for a downstream use case (ML model input, visualization, or another project artifact).
N	Fewer than 2 queries, or queries use advanced features but without a clear role in the project, or the purpose stated is trivial (“to demonstrate a window function”).
U	Queries are simple <code>SELECT...WHERE...</code> with no advanced features, or have no identifiable purpose in the project.

- **Performance tuning.** Choose 2-3 queries that perform poorly and add indexes or otherwise modify the queries or database to improve performance. Your changes must result in at least a 20% reduction in execution time. In your Artifact Self-Assessment, include relevant “before and after” snippets from the query plan **and** timings or other evidence of the speed up. (If you chose the **Advanced technology** option, you may provide alternative evidence appropriate to your database.)

Grade	Description
E	All S-level requirements met, with exceptional execution: deeper analysis of why the changes worked (or didn’t), more queries, or unusually thoughtful tuning (e.g., multi-column indexes chosen from query plan analysis, query rewrites that change join strategy).
S	For each of 2–3 queries: runs the query at least 5 times before and 5 times after the change and reports mean and standard deviation of execution time; documents each tuning change made (index creation, query rewrite, etc.) and its measured impact; shows <code>EXPLAIN ANALYZE</code> output before and after and describes the difference in the query plan. At least 20% reduction in mean execution time overall. If a change produced no improvement, the group explains why (e.g., plan did not change, data too small for index to matter).
N	Measurements are reported but without 5+ runs or mean/stddev; query plans are not shown or not compared; or the 20% threshold is not met and the lack of improvement is not explained.
U	No before/after measurements, no query plans, or no documented changes.

- **Performance comparison.** If using the **Advanced technology** option, perform a comparison of performance between your chosen database and PostgreSQL. Provide at least 3 carefully designed benchmarks (data loading, query execution, or other tasks) which you execute on both systems. In your Artifact Self-Assessment, compare the results using statistical analysis techniques, including at least one visualization. Provide a detailed analysis of why PostgreSQL or the other system performs better in each case.

Grade	Description
E	All S-level requirements met, with unusual depth: benchmarks deliberately stress features where the two systems should diverge, hypotheses are verified with additional drill-down (e.g., query plan inspection, system-level metrics), and discussion draws meaningful conclusions about when to prefer each system.
S	3 benchmarks that are <i>sufficiently different</i> to exercise distinct operators, features, or workloads (not 3 variants of the same query). For each benchmark: show how the task is expressed in each system (SQL for Postgres, equivalent query/operation in the other); run at least 5 times per system and report mean and standard deviation; include an appropriate visualization of the results; state a hypothesis for <i>why</i> the observed performance difference arose (e.g., storage format, index structure, parallelism).
N	Fewer than 3 benchmarks, or benchmarks are too similar, or measurements lack 5+ runs / mean / stddev, or visualizations are unclear, or no hypothesis is given for the results.
U	Benchmarks not actually run on both systems, or no quantitative comparison presented.

- **Statistical and machine learning analyses.** Provide 2-3 careful analyses of your data that reveal something of interest that is not obvious from a casual examination of the data. Simple statistics and 2-variable correlations are not interesting analyses, unless accompanied by significant work to transform the data and derive the variables of interest. At least one analysis must be performed in SQL or enabled by SQL (e.g., through aggregation). Include properly labeled and formatted visualizations where appropriate.

Grade	Description
E	All S-level requirements met, with exceptional rigor or insight: e.g., principled model comparison, ablations, well-designed cross-validation, or interpretation that connects results back to real-world implications in a non-obvious way.
S	For each analysis: (1) motivates the model choice – explains why this model's inductive bias is appropriate for the task; (2) performs hyperparameter tuning with evidence that multiple configurations were explored and evaluated; (3) motivates the evaluation – explains why the chosen metrics are appropriate for the task and interprets the metrics to conclude how well the model supports the task; (4) clearly explains the role of the database : e.g., data stored in SQL, feature engineering performed in SQL, results stored in SQL for downstream analysis. At least one analysis must be performed in SQL or enabled by SQL.
N	Analysis is present but missing one or more of: model motivation, hyperparameter exploration, metric justification, or clear DB role.
U	Trivial analysis (basic statistics, simple correlations with no transformation), or no SQL involvement, or no interpretation of results.

- **Data visualization.** These are **not** visualizations of your **Statistical and machine learning analyses**, if you are doing them. These are other, creative ways of visualizing your data or summaries of your data, such as Sankey diagrams, dendograms, etc. No sunburst diagrams allowed. Include the visualization(s) (properly labeled and formatted) and a discussion of the significant features shown in the visualization – what is this visualization telling us? Your selected visualizations must be used effectively and correctly – do not simply find a way to conform your data to the requirements for the plot type. For example, radial plots such as sunburst diagrams should illustrate proportional, part-to-whole relationships, while Sankey diagrams should represent flows. All labels must be readable. Check out this site for some guidance and inspiration: <https://guides.lib.utexas.edu/data-visualization/examples>.

Grade	Description
E	All S-level requirements met, and the visualization work goes well beyond a standard chart library: e.g., develops custom visualization software, implements an idea from a research paper, or pursues a genuinely unique visualization that is well suited to the task, has a clear purpose, and reveals something a standard plot could not.
S	Each visualization is justified – the plot type is appropriate to the underlying data relationship (e.g., Sankey for flows, dendrogram for hierarchical clustering) and is not being forced to match the data. Axes, legends, and color choices are clearly labeled, readable, and visually distinguishable. The discussion identifies what the visualization reveals about the data.
N	Plot type is mismatched to the data relationship, labels are missing or unreadable, or the discussion does not explain what the visualization shows.
U	Requirements not met, or used a sunburst diagram.

- **Software.** If you chose the **Software development** application, provide your source code either as a link (in your Artifact Self-Assessment) to a public repository or as additional files in your submission. Your Artifact Self-Assessment should describe the functions of your application that query, add, modify, and remove data, including examples of the SQL (or other query language) used in your application (properly formatted with a fixed-width font in your submission text, not a screenshot!). Provide screenshots or transcripts of the application in action to illustrate each function (but keep it concise - ellide long outputs or transcripts).

Grade	Description
E	All S-level requirements met, with exceptional polish: e.g., thoughtful UX, careful error handling, authentication/authorization where appropriate, automated tests, or a deployment story.
S	Source code is fully functional and implements all four CRUD capabilities (query, add, modify, remove) against the database. Uses parameterized queries / prepared statements to prevent SQL injection. The Artifact Self-Assessment describes each function's purpose with example SQL (formatted as text, not a screenshot) and shows screenshots or transcripts of each function in action.
N	Source code runs but one of the CRUD capabilities is missing or incomplete; or SQL injection protection is absent; or the Artifact Self-Assessment does not document each function with examples.
U	Source code is not functional.

- **Evaluation of technology.** If you chose the **Advanced technology** application, provide a summary of the key features of the database technology you selected and how it differs from an RDBMS; provide a compelling demonstration of the advantages or trade-offs of using the new technology. This could be descriptions of queries that are made possible or much easier with the new technology, deployment considerations that are possible with the new technology, or other interesting factors which are **not** related to performance. For performance related comparisons, please see the relevant output (earlier in this section).

Grade	Description
E	All S-level requirements met, with exceptional depth: both quantitative and qualitative evidence, real-world scenarios where the chosen technology is clearly preferable, and thoughtful discussion of where it is <i>not</i> a good fit.
S	Summarizes the key features of the selected technology and how it differs from an RDBMS. Demonstrates non-performance advantages or trade-offs with concrete evidence – e.g., a query or operation that is substantially easier or only possible in the new system, a deployment consideration, or a data-model feature – supported by experimentation or cited research. Discusses both advantages and disadvantages.
N	Lists some differences between the technology and an RDBMS but with little evidence, or only anecdotal discussion; trade-offs are not substantiated.
U	No meaningful discussion of trade-offs, or discussion is purely performance-oriented (belongs in the performance comparison output).

4.3.1 Grading

Each output is graded on the S/N/U rubric specified in that output’s description. For each output you attempt, you must also provide the artifacts required in the output’s description (e.g., the SQL script for data loading and cleaning, the source code for software). Credit is only given for non-trivial artifacts that are genuinely relevant to the project; you cannot rack up outputs by turning in meaningless artifacts across many categories.

5 Artifact Self-Assessment

There is **no separate final report** for this project. Instead, you will submit an Artifact Self-Assessment: we will provide a template containing the rubric tables for Components 1, 2, and each Component 3 output you chose. Your job is to self-assess and justify your claim, not to write a long paper.

For each applicable rubric, your group will:

- Check the E/S/N/U box corresponding to the grade you believe you have earned.
- Write a short justification (1–2 sentences) explaining *why* your work is at that level, referencing specific artifacts (queries, figures, code, etc.).
- Attach the supporting artifacts required by the output’s description (e.g., the SQL script for data cleaning, the query plans for performance tuning, a link to the source code repository for software).

The template will also include:

- A list of group members.
- An individual contribution statement from **each** group member, listing all members with the percentage of overall project effort they contributed (including the writer). Each member submits their own contribution statement to Canvas; one member submits the filled-in template plus auxiliary files (source code, scripts) as a .zip archive.

Instructors will grade against the same rubrics, so an honest and accurate self-assessment makes the process easier for everyone – overclaiming will be corrected down.

6 Presentation

Your group will give a ~5-minute “lightning talk” during the last week of class, depending on the number of groups presenting that day. Sign-up sheets for each presentation day will be posted in advance. You may present as a group or select a subset of your group to do the presentation. Your talk needs to do the following:

- Communicate **Component 1** (dataset) and **Component 2** (application/technology) clearly enough that the audience can judge whether they meet the rubric.

- Present the **Component 3 outputs** you are presenting and explain how to interpret the evidence you show.
- For each component/output you present, show enough concrete evidence that the class can judge **S-level execution** against the same rubric you used in the Artifact Self-Assessment.

Presentations are graded in part through **peer grading**: every student in the class will have a copy of the same rubrics and will check off E/S/N/U for the components and outputs the presenting group covers, based on what is shown in the slides and talk. These peer rubrics will be submitted after each presentation and factor into the presentation grade alongside the instructor's evaluation.

7 Grading

Your final project will receive two grades: one for the work itself – what did you do, and at what quality, and one for the presentation of your work – the lightning talk.

7.1 Project work

Your project-work grade aggregates the grades on Components 1, 2, and 3 using the rubrics above.

Grade	Description
E	All S-level requirements met across Components 1, 2, and 3, with at least one output achieving an E and no component below S. The project as a whole demonstrates creativity, quality of execution, and synthesis beyond the baseline requirements.
S	Components 1 and 2 each at S or better; at least 4 outputs in Component 3 each at S or better. Individual contribution is sufficient and consistent with the group's contribution statements.
N	Components 1 or 2 below S, or fewer than 4 Component 3 outputs at S, or individual contribution is clearly below the group average.
U	Multiple components at U, or essential required artifacts (source code, SQL scripts, dataset citations) are missing.

Your individual score depends on this rubric **and** on your contribution to the project. If we feel your contribution (based on the contribution statements of everyone in the group) is insufficient, we will reduce your individual grade accordingly.

7.2 Presentation

The presentation grade covers the lightning talk.

Grade	Description
E	All S-level requirements met, with exceptional communication: the lightning talk is polished, clearly motivated, and especially effective at presenting convincing evidence for the components and outputs being claimed.
S	Lightning talk clearly communicates Component 1, Component 2, and the Component 3 outputs being presented. The slides and verbal explanation provide enough specific evidence that peers can reasonably score the same rubrics live, including judging whether the presented work demonstrates S-level execution.
N	Lightning talk is missing one or more required pieces (Component 1, Component 2, or the claimed outputs), or the evidence shown is too vague or incomplete for peers to judge the rubric confidently.
U	No lightning talk given, or the talk is so incomplete that peers cannot meaningfully score the same rubrics from what is shown.

Some resources for finding a dataset

- Kaggle: <https://www.kaggle.com/datasets>

- UC Irvine ML repo <https://archive.ics.uci.edu/ml/index.php>
- Governmental data repositories such as:
 - <https://data.gov>
 - <https://www.census.gov>
 - <https://data.colorado.gov>
 - <https://opendata-geospatialdenver.hub.arcgis.com>
- Scientific data repositories like <https://zenodo.org> and <https://dataverse.harvard.edu>
- Other data repositories:
 - <https://www.responsible-datasets-in-context.com>,
 - <https://github.com/awesomedata/awesome-public-datasets>,
 - <https://www.gapminder.org/data/>,
 - <https://www.pewresearch.org/internet/datasets/>
 - <https://data.fivethirtyeight.com>